Amazon AI Cyber Security Hackathon, Round 1 submission

Track 02: AI-Driven Scam Pattern Recognition

ScriptBreak

Break the scam script before the PIN.

ScriptBreak is an on-device layer for UPI apps that spots payments matching a coached-scam pattern, shows the plain facts before the PIN, and brings in family with one tap.

Reads no call audio, messages or contact list

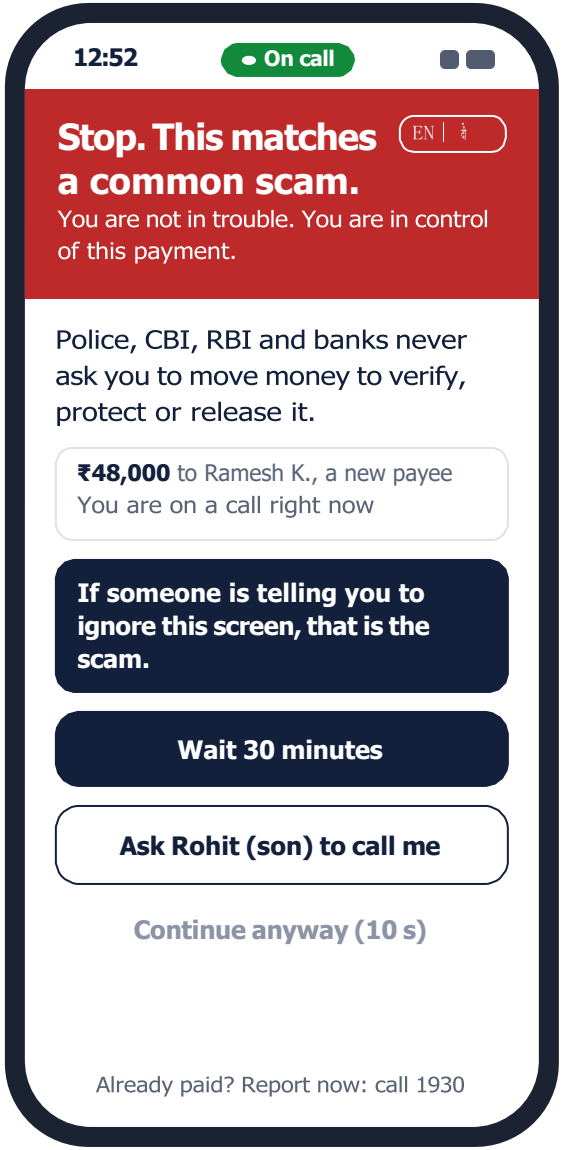
Never blocks a payment: the person decides

AI proposes candidate rules from redacted reports; open rules decide at the PIN

People approve every rule before it reaches a phone

Team RUNTIME, Chitkara University: Sambhav Jain, Sehjal Saxena, Deepak Rawat

Problem fit Slides 2–3	**Architecture and logic** Slides 6–7, pages 11–13	**Value and differentiation** Slides 3–5	**Feasibility and 48 hours** Slides 8, 10
Safety and responsible AI Slide 9, page 13	**Adoption and scale** Slide 10, page 12	**Evidence we ran** Slides 7–8, appendix A–C	**Disclosures and sources** Appendix E



The money is lost in the seconds before the PIN, while the scammer is still on the line

After the PIN, money can move in seconds, and recovery is hard: of ₹36,448 crore reported lost on the national cybercrime portal by February 2025, ₹4,381 crore was frozen, but only ₹60.52 crore had been returned.

Mr. Sharma, 64, retired teacher, Pune

Synthetic scenario built from public advisories and reporting

11:40	A video call from a "CBI officer" in uniform. Faces and voices can be AI-generated, making the caller look and sound convincing.
11:55	He is told he is under "digital arrest": stay on camera, tell no one, not even family.
12:48	"Transfer ₹48,000 to this verification account. It will be returned after checking."
12:50	**He opens his UPI app, enters a new payee and types ₹48,000. Nothing on the screen connects the call to the payment.**

₹22,495 crore financial fraud reported on India's national cybercrime portal in 2025, from 24 lakh complaints

21× growth in reported "digital arrest" losses in two years, from ₹91 crore in 2022 to ₹1,935 crore in 2024

4× Senior citizens lost about ₹3.9 lakh per complaint in 2025, about four times the average of ₹0.94 lakh

Primary user and harmful moment

Anyone paying by UPI while a stranger coaches them live; highest risk for people over 55 and first-time digital users, in the seconds between typing the amount and entering the PIN. Also involved: a trusted family member and the bank's fraud team.

Threat pattern (in scope)

Push payments made while a stranger guides the victim on a voice or video call or over screen-share: digital arrest, fake customer care, fake KYC or bank official, courier scams, and AI voice clones of relatives.

Out of scope for now

Account takeover, fake shopping sites and card fraud. Scams coached only over chat are partly covered by a rule our AI loop proposed and we approved (slide 7). We show where we fail, not just where we work.

The payment looks genuine; the session around it doesn't. Most existing defences see only one side of the scam.

Existing defence	What it does	What it misses at the moment of payment
Bank transaction monitoring	Scores each payment for unusual patterns	The victim pays with their own PIN, so it can look genuine. It cannot see the call.
2025 UPI changes	P2P collect requests ended; verified recipient names shown	Push payments remain. Victims may be told the stranger's name is "the officer's account".
Android scam alerts (pilot)	Warns when you open a partner finance app during a call with an unknown number	Fires when the app opens, before a payment exists, so it can't say which payment is risky or why. Partner apps on Android only.
On-device call scam detection	Analyses call speech for scam patterns	Pixel 9 and later only, off by default. AI can generate endless script variations.
Mule-account detection	Finds receiving accounts used by fraudsters (MuleHunter.AI, I4C Suspect Registry)	Works on the receiving side. A fresh account isn't known yet, and the victim gets no explanation in the moment.
Awareness campaigns	Teach that "digital arrest" isn't real	The scammer is on the line, talking the victim past every generic warning.

Each defence sees one piece of the scam.

AI makes every word, voice and face of a scam cheap to change. It can't easily hide the shape of a coached payment: someone guiding the victim live, an account never paid before, an amount out of character.

ScriptBreak joins those facts at the one moment that matters, just before the PIN, inside the app the victim already uses. It complements the defences on the left; it doesn't replace them.

ScriptBreak adds one screen to a coached payment and nothing to a normal one

Today

1. Video call from a "CBI officer". 70 minutes of threats; told to tell no one.
2. Opens his UPI app and enters the "verification account" as a new payee.
3. Types ₹48,000. A generic fraud banner appears.
4. Scammer: "That banner is routine. Continue."
5. Enters his PIN. ₹48,000 leaves instantly.
6. Tells his son hours later. The money has moved on through mule accounts.

With ScriptBreak

1. Same call, same app, same new payee.
2. ScriptBreak sees: a call is active, he has never paid this account, and ₹48,000 is 74× his usual. Level: Pause.
3. Facts first: "This sends ₹48,000 out of your account to Ramesh K. You have never paid this person."
4. "Were you told this payment is needed to verify, protect or release your money?" He taps Yes. Level: Critical.
5. He taps "Ask Rohit (son) to call me". His son's phone gets a short alert.
6. In this scenario, Rohit calls, Mr. Sharma hangs up on the scammer, and no money leaves.

The payer

Sees what this payment actually does, in one sentence, and stays in control of the decision.

The family

Police advice: the scam "dies the moment isolation breaks". One tap breaks it, and a real relative can verify whether the request is genuine.

The bank

Can help prevent coached losses and disputes, and gets consented, redacted reports that can reveal new scam patterns.

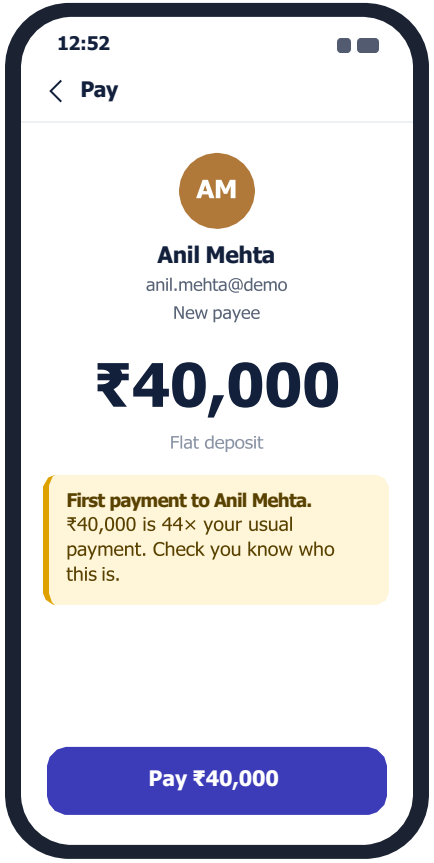
Four levels are designed to keep alert fatigue low



None

Nothing extra

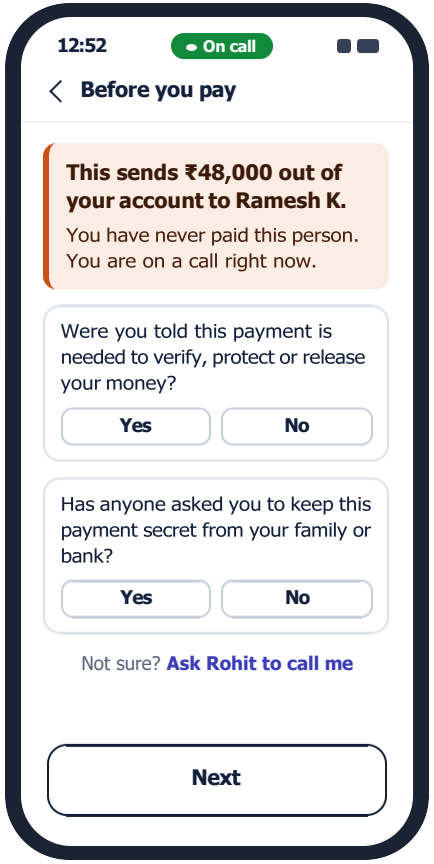
Most payments. A payee first paid more than a week ago, with no earlier warning, never counts as new.



Nudge

One line, no extra tap

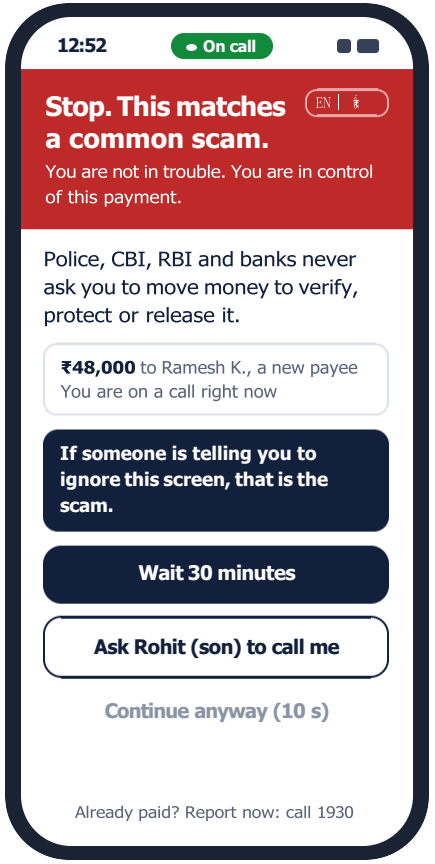
New payee plus a call or an unusual amount, ₹1,000 or more. Shown on 3.3% of payments in our synthetic stress test, vs 20.8% for a blanket new-payee notice.



Pause

Facts first, then two questions

Live guidance + new payee + unusual amount; screen-share or remote access + new payee; or a learned rule.



Critical

Warning + family option + 10-second confirmation delay

A Pause plus a "Yes", a warned or bank-flagged payee, or screen-share with an unusual amount. "Continue anyway" stays available.

At the PIN, open rules read the shape of a coached payment, not the words of the script

C Live guidance
a call (now or in the last 30 min) or screen-share

+

P New payee
never paid, or first paid under 7 days ago

+

A Unusual amount
for this person, or added up over 24 hours

→

Pause
facts first, then two questions

→

Critical
if the person answers "Yes"

No single signal interrupts a payment, except a bank-flagged payee. **Answers make coaching harder:** they can raise the level, never lower it. **Synthetic case S01:** on a call + never paid Ramesh K. + ₹48,000 (74× his usual) → Pause; "Yes" to Q1 → Critical.

What each signal means (read only while the app is open)

C Live guidance
A phone or internet call is active during the payment, or was seen while this app was open in the last 30 minutes. Strong: screen sharing, screen recording or a remote-access app.
Not used: call length, caller's number, audio (page 12). Simulated in the prototype.

P New payee
A person or a merchant; only allow-listed large billers are exempt. Strong: paid after an earlier Pause or Critical warning (a Nudge doesn't count), or bank-flagged.

A Unusual amount
5× the person's own 90-day median and ₹2,000+; ₹5,000+ with little history. Payments to new payees in the last 24 hours are added up, so splitting is caught once the running total crosses the threshold.

? Two questions, only at Pause
Q1 "Were you told this payment is needed to verify, protect or release your money?" Q2 "Has anyone asked you to keep this payment secret from your family or bank?"

Decision table, rule pack v0.3

None	No rule matches: most payments.
Nudge	New payee + (C or A), ₹1,000 or more; or a warned payee without C: one line, no tap.
Pause	**R1** C + P + A. **R2** screen-share or remote access + new payee. **R3** C + warned or bank-flagged payee. **R4** bank-flagged payee. (R3, R4 always end in Critical.) **R5, R6** learned rules (slide 7).
Critical	A Pause plus a "Yes", a warned or bank-flagged payee, or screen-share or remote access with an unusual amount.

Why rules at the PIN, not a model score? The brief warns that "a detector score alone is not an explanation or proof". Our rules produce the exact reasons shown on screen, and a bank can audit them. AI works upstream: it proposes rules for what they miss, and people approve them (slide 7).

AI proposes, people approve: the loop recovered two scam patterns our rules missed

Synthetic scam reports with no scam-type labels.

1 Reports
1,200 synthetic scam reports
Consented and redacted: events only, no names, numbers or text.

2 Misses
449 not paused
The reports the current rules did not catch.

3 Cluster (AI)
7 patterns
Misses grouped by the events they share.

4 Propose (AI)
One rule per pattern
Allowed at most 0.5 extra false alarms per 1,000.

5 People decide
Approve, test or reject
Two approvals (in this run: our team, applying the written policy in appendix C); then shadow mode before going live.

R5 approved (synthetic run)

"Sent by mistake"[6]: money returned to a different account

Rule: same amount as money received from an unknown sender + new payee, not paying back that sender

100%	+0.05	17% → 100%
of the pattern caught	false alarms per 1,000	flagged in a synthetic held-out test

R6 approved (synthetic run)

Task scam over chat: 3rd new account within an hour

Rule: left the app and came back + 3rd new payee within an hour + new payee + payee ID pasted

93%	+0.02	5% → 27%
of the pattern caught	false alarms per 1,000	flagged in a synthetic held-out test

Not approved. Four proposals were rejected for adding too many false alarms; one caught too little (55%) and stays in shadow mode. Catching "hang up, then pay" would add 2.8 per 1,000 in the synthetic backtest, so it stays a stated limit (page 13). Full run: appendix C.

What it can and cannot do. It finds new combinations of events the phone already records. A new kind of event needs a new collector, built and reviewed by people. These reports are synthetic; the real test is shadow mode.

What we tested on synthetic data, and the one assumption that decides false alarms

28 of 28

Behaviour checks pass

14 of 16 synthetic scam cases flagged; 2 of 10 genuine lookalikes interrupted, as designed (appendix A).

68% vs 39%

Synthetic scam sessions flagged, at a similar false-alarm rate

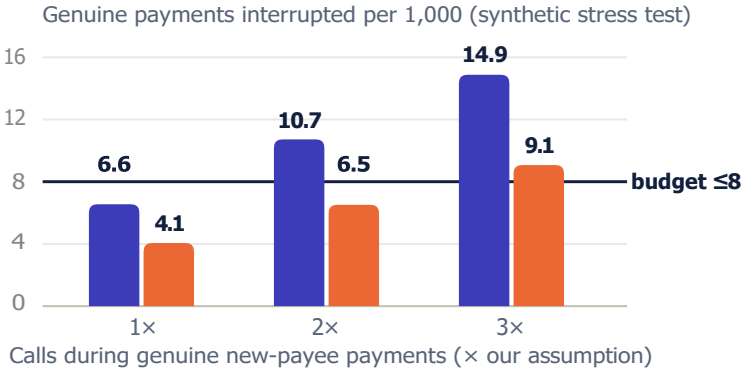
ScriptBreak vs an amount-only rule (first payment to a person ≥ ₹7,500): 6.6 vs 5.7 genuine payments interrupted per 1,000. An amount-only rule needs 29 per 1,000 to flag as many.

6.6

Genuine payments interrupted per 1,000 (synthetic stress test)

Interruption budget ≤8 per 1,000: at that rate, someone making 40 payments a month would see about one Pause every three months.

The assumption that matters most



Rule pack v0.3

Pre-planned fallback: ₹5,000 amount floor

Budget reached at 1.35× our assumption; with the fallback, at 2.6×. Shadow mode measures this number first.

What these tests can and cannot show

Synthetic tests validate the logic, not real-world prevalence. The first production gate is the interruption rate measured in bank shadow mode.

Key assumption	How we validate it
A1 Apps can read call and screen-share state while open	Documented OS interfaces (page 12); probe app after the finale
A2 Genuine new-payee payments during calls are at most 1.35× our assumption	8-week shadow mode; if higher, switch on the tested ₹5,000 fallback

Scam averages cover six in-scope workflows (appendix B), excluding "sent by mistake", which the learned rule catches 100%. Data: synthetic only, from public advisories. Three more assumptions: page 13.

Safe when it is right, low-cost when it is wrong

Data

Used, on the phone only

- Call or screen-share active
- Payee history, money received
- Amount vs your own median
- Pasted payee ID, app switches
- Your two answers

Never collected

- Call audio, caller number, call length, call log
- Message text; contact list (only the one trusted contact you enter)
- Location; clipboard contents

Read only while the app is open.

When we are wrong

- **False alarm:** one extra screen; "Continue anyway" always works.
- **Missed scam:** one-tap report packet for the 1930 helpline, so the bank can try to freeze funds quickly.
- **Engine error:** the payment goes through; we fail open.

Who can act

- **The person** always decides.
- **Family** can call, but can't see the account or stop the payment.
- **Bank analysts** see only redacted reports and approve rules.
- **Nobody is accused:** the screen says "This matches a common scam."

Misuse and bias

- Coached answers can't lower the level.
- New trusted contact: OTP and a 24-hour delay.
- Learned rules: two analysts, budget and all checks.
- Interruption rates compared across age groups and first-time users each release.

"If someone is telling you to ignore this screen, that is the scam."

The last line of every Critical warning, written against the scammer's main counter-move: coaching the victim past the warning.

Ships as a small SDK inside apps people already use. In 48 hours, judges watch it catch a simulated scam live.

Prototype: OS signals simulated. **Production:** only host-app or OS-authorised signals.

Deployment pathway

The host app calls ScriptBreak after the payee and amount are entered, before the PIN pad:

```
result = ScriptBreak.assess(payment, device)
// on the phone, no network call, target < 50 ms
// returns { level, reasons[], screenCopy, rulePack }
```

Owner: the bank's or payment app's fraud-risk team; the customer-experience team owns wording and languages.

Phase 0	**Shadow mode, 8 weeks:** levels computed but not shown; interruption rate measured.
Phase 1	**All four levels and Ask family,** for an opt-in group of customers over 55.
Phase 2	**All customers.**
Phase 3	**Learning loop** live; signed rule packs shared across partner banks.

Each phase needs the interruption budget met and the bias check passed. Friction: none for most payments. Pause: target under 30 seconds; Critical adds a 10-second confirmation delay. Both to be measured in the hallway test (A3).

48-hour MVP scope

Must

1. Mock UPI app with a scenario panel
2. Engine and all 28 checks in TypeScript
3. The four screens
4. "Ask family" to a second phone
5. Live scoreboard

Should

6. Hindi Critical screen
7. Report packet
8. Analyst view of the discovery loop (already runs in Python)

Stretch

9. Hindi on every screen
10. Live analyst approve button

Not in 48 hours: native SDK, real OS signals.

Dependencies: one free realtime relay for the family alert (local WebSocket fallback); no paid APIs; no external data.

0–8 h	Repo and live URL (Sehjal Saxena). Engine and checks in TypeScript (Sambhav Jain). Generator port (Deepak Rawat). **Checkpoint: 28 of 28 checks pass; app live.**
8–24 h	Screens and scenario panel (Sehjal Saxena). Engine wired to the UI (Sambhav Jain). Family alert relay (Deepak Rawat). **Checkpoint: full flow on two phones.**
24–36 h	Scoreboard (Sehjal Saxena, Deepak Rawat). Should items: Hindi screen, report packet, analyst view (all). **Checkpoint: every Must item demoable.**
36–48 h	Stretch only if all checkpoints are green; rehearse, record a backup video, freeze. **Checkpoint: demo runs 5 times without a fault.**

Demo: each step proves a claim

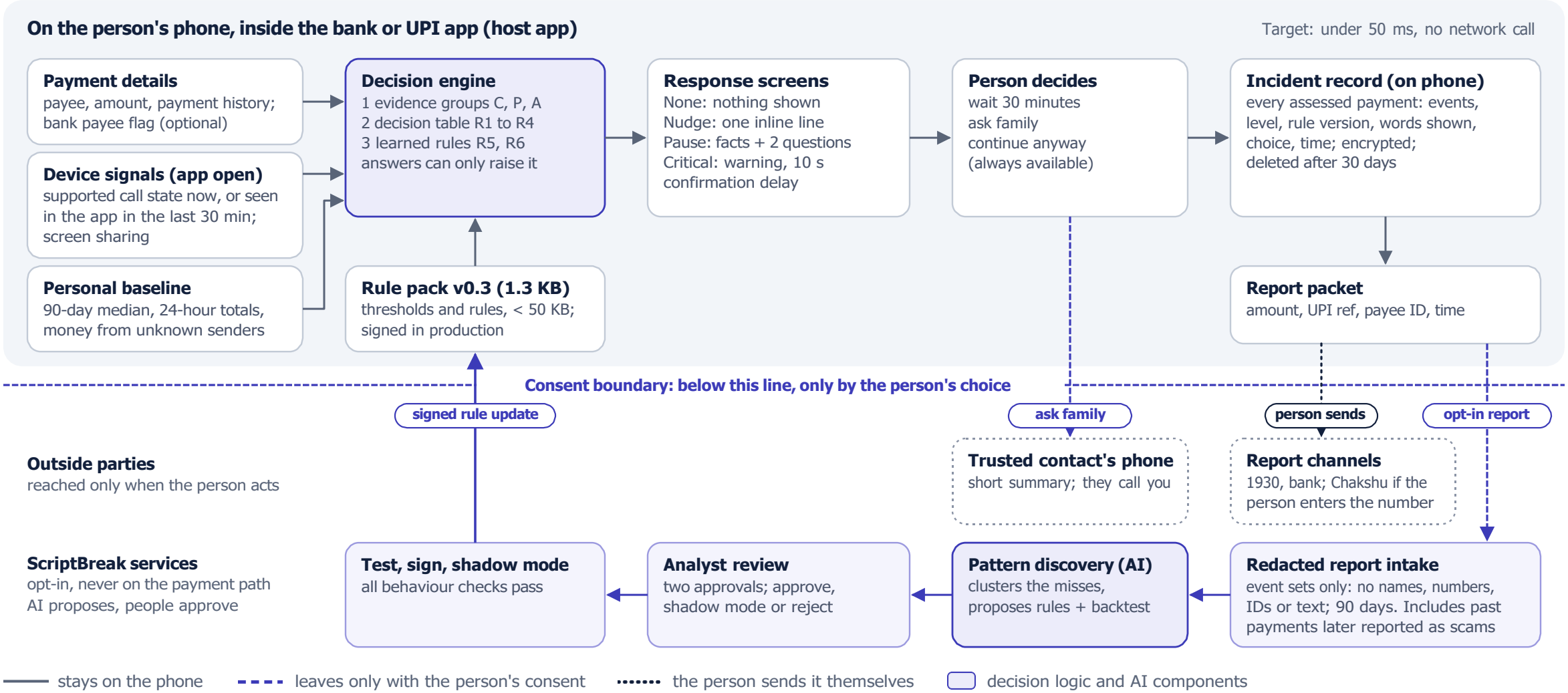
1. Digital-arrest call: Pause, then Critical; the family phone buzzes (detection, escalation)
2. ₹18,000 rent to the usual landlord: nothing shown (no alert fatigue)
3. Coached "No, No": still paused (answers can't lower the level)
4. "Sent by mistake" return: v0.2 nudges, v0.3 pauses (learning loop)
5. Scoreboard: all 28 checks and interruptions per 1,000, live (judges can verify)

Sehjal Saxena: Frontend and UI, deployment, some backend

Sambhav Jain: Backend, decision engine, rules, analyst view

Deepak Rawat: Backend, generator, family alert, scoreboard

Decisions happen on the phone. Data leaves it only when the person chooses.



Interfaces and outputs: `assess()` returns the level and reasons to the host app; the screens show the explanation; an encrypted incident record stays on the phone. Only if the person chooses: a family alert, a report packet and a redacted report. Rule packs (signed in production) flow back to phones.

Every input has a reason, a consent path and a documented way for an app to get it

Input	Why	Android	iOS	Consent; never collected
Supported call / communication state	Coaching happens on calls	Audio mode "in call" or "in communication". Voice-chat apps can set it too; that counts in the false-alarm budget	CallKit call observer (phone calls, and VoIP apps that use CallKit)	Explained at onboarding; never caller, length or audio
Call seen in app, last 30 min	"Hang up, then pay"	Works only if the app was opened during the call: it stores when it last saw one. Nothing runs in the background		Same; never call history
Screen sharing or remote access	Remote-control scams	Android 15+: screen-recording callback. Android 11–14: known remote-access apps only	Screen capture state	Host app's device check; never screen contents
Payee history, money received	New accounts; "sent by mistake"	The host app's own payment records, on the phone		Part of using the app
Amount baseline, 24-hour totals	Unusual for this person	Computed on the phone		History never uploaded
Pasted payee ID, app switches	Learned rule R6	The app's own text field and screen lifecycle		Never clipboard contents
Two answers	Raise the level only	The person, at Pause only		Optional; no free text
Bank payee flag	Reported accounts	Host bank feed, e.g. I4C Suspect Registry[11]		Bank-controlled feed
Trusted contact	Family escalation	Entered by the person		OTP; 24-hour delay on change

Not used: call log and caller number (Google Play restricts call-log access[17]; iOS doesn't expose them), call length, contact list, message text. **Availability** depends on OS version, host-app permissions, platform policy and bank/PSP integration. The prototype simulates all device signals.

From signal to decision to evidence

1. Payee and amount entered; the host app calls `assess()` on the phone.
2. Rules give a level and its reasons; the rule pack version is recorded.
3. The screen matching the level is shown (slide 5).
4. The person chooses: wait 30 minutes (closes this payment without paying), ask family, or continue. Never overridden.
5. **Incident record** (the evidence) on the phone for every assessed payment: events, level, rule version, words shown, choice, time. No content. Deleted after 30 days.
6. **Escalation, person-initiated:** family alert; report packet to 1930, cybercrime.gov.in or the bank; if the person chooses, they can enter the caller's number themselves on Chakshu.
7. **Learning, opt-in:** redacted events only (no names, numbers, IDs or text), kept 90 days, feed the discovery loop.

Designed for a scammer who adapts, with every remaining gap named

Attack	Mitigation	Residual risk
Coaching: "answer No", "the warning is routine"	Answers only raise the level; the Critical screen names that exact line as the scam	Victim may still continue
Hang up, then pay	Call seen in the app in the last 30 min; runs alongside the bank's existing amount rules, which catch more of this case in our synthetic tests	Whole payment after the call: Nudge only (S12). A later repeat to a warned payee without a call also gets only a Nudge.
Splitting amounts	New-payee payments in the last 24 hours added up; 3rd new payee in an hour. Thresholds are per person and never shown.	First small payment: Nudge (S08)
Merchant accounts	Merchants not exempt; only allow-listed large billers	+1.6 false alarms per 1,000 (synthetic ablation, appendix B)
AI voice clone of a relative	Secrecy question (Q2), and "Ask family" on both Pause and Critical, which reaches the real relative	A victim who trusts the voice may still pay; the facts still show
Poisoned reports	Rate limits, attestation, two analysts, behaviour checks, shadow mode	Slow drift; reviewed each release
Family alert hijack, tampering, leaks	OTP and 24-hour delay; signed rule packs; redaction on the phone; role-based access, audit logs	Depends on host-app security
App hopping: "use your other UPI app"	Small SDK; rule packs shared across partner banks (Phase 3)	Residual until most UPI apps include it

Failure policy: if anything in ScriptBreak fails, the payment goes through and the error is logged. A ScriptBreak failure can never block the payment.

Performance targets (to measure)

Decision time	**Target under 50 ms**, no network
Offline	**Works;** family alert queued
Rule pack	**Under 50 KB** (v0.3: 1.3 KB); signing planned
Battery / background	No background monitoring; read **only while the app is open**
Interruption budget	**≤8 per 1,000** genuine payments
Retention	Phone incident record 30 days; redacted reports 90 days

Validation approach

Done	On synthetic data: 28 behaviour checks, stress test, discovery loop with held-out test.
Finale	Two-phone demo; checks and scoreboard live in the build.
Pilot	Probe app (A1); shadow mode (A2). **A3** people answer under pressure: hallway test, 10 people, 3 over 55. **A4** family responds in minutes: pilot logs. **A5** reports are honest: poisoning tests.

Behaviour checks: 28 synthetic cases, expected against observed

v0.2: hand-written rules; v0.3: plus learned R5, R6 (`run_cases.py`). **a** Known limit (page 13). **b** Caught only after a learned rule. **c** Accepted false alarm. *Amount rule: first payment to a person, ₹10,000+. Warned payee: paid after an earlier Pause or Critical warning. Q1: "Were you told this payment is needed to verify, protect or release your money?" Q2: "Has anyone asked you to keep this payment secret from your family or bank?"

ID	Case	Signals present	Expected	v0.2	v0.3	Match (v0.3)	Amount rule*
In-scope scams (coached during a call or screen-share)							
S01	Digital arrest; victim confirms the "verification" story	on a call, new payee, ₹48,000 (74× usual), Q1 Yes	Critical	Critical	Critical	Yes	Warns
S02	Digital arrest; coached victim answers "No" to both questions	on a call, new payee, ₹48,000 (74× usual), Q1 No, Q2 No	Pause	Pause	Pause	Yes	Warns
S03	Digital arrest, day 2: same account again after a warning	on a call, warned payee, ₹1,50,000 (231× usual)	Critical	Critical	Critical	Yes	–
S04	Fake customer care: pay ₹4,000 to "release" a refund	on a call, new payee, ₹4,000 (11× usual), Q1 Yes	Critical	Critical	Critical	Yes	–
S05	Fake KYC update on screen-share; ₹9,999 kept under ₹10,000	on a call, screen shared, new payee, ₹9,999 (12× usual)	Critical	Critical	Critical	Yes	–
S06	Split payments: 3rd ₹1,900 transfer to new accounts in an hour	on a call, new payee, ₹1,900 (1.6× usual), 3rd new payee in 1 h	Pause	Pause	Pause	Yes	–
S07	"Parcel with drugs" courier call; told to keep it secret	on a call, new payee, ₹25,000 (12× usual), Q2 Yes	Critical	Critical	Critical	Yes	Warns
S08	Fake bank official: first ₹3,500 "to secure your account"[a]	on a call, new payee, ₹3,500 (2× usual)	Nudge	Nudge	Nudge	Yes	–
S09	Same official asks for a third ₹3,500 to the same account	on a call, payee first paid in last 24 h, ₹3,500 (2× usual), ₹7,000 to new payees in 24 h	Pause	Pause	Pause	Yes	–
S10	₹1,999 sent a 5th time to one account; the 4th was paused	on a call, warned payee, ₹1,999 (1.7× usual), ₹7,996 to new payees in 24 h	Critical	Critical	Critical	Yes	–
S11	Told to hang up and pay; had checked balance during the call	call in app, last 30 min, new payee, ₹48,000 (74× usual), ID pasted	Pause	Pause	Pause	Yes	Warns
S12	Told to hang up and pay; whole payment made after the call[a]	new payee, ₹48,000 (74× usual), ID pasted, left app	Nudge	Nudge	Nudge	Yes	Warns
S13	First-time digital user (no history) in a digital-arrest call	on a call, new payee, ₹20,000 (little history)	Pause	Pause	Pause	Yes	Warns
S14	Scammer collects through a merchant account during the call	on a call, new merchant, ₹30,000 (33× usual)	Pause	Pause	Pause	Yes	–
S15	"Sent by mistake" scam: return ₹5,000 to a different account[b]	on a call, new payee, ₹5,000 (2× usual), ₹5,000 from an unknown sender	Pause	Nudge	Pause	Yes	–
S16	AI voice clone of a son: "₹30,000 now, don't tell Dad"	on a call, new payee, ₹30,000 (20× usual), Q1 No, Q2 Yes	Critical	Critical	Critical	Yes	Warns
Boundary: coached over chat only							
B01	Task scam coached over chat only, no call	new payee, ₹5,000 (10× usual), 2nd new payee in 1 h, ID pasted	Nudge	Nudge	Nudge	Yes	–
B02	Task scam over chat: 3rd new account within an hour[b]	new payee, ₹3,000 (5× usual), ₹2,500 to new payees in 24 h, 3rd new payee in 1 h, ID pasted, left app	Pause	Nudge	Pause	Yes	–
Genuine lookalikes							
L01	₹350 at the usual kirana store	known merchant, ₹350 (below usual)	None	None	None	Yes	–
L02	Monthly rent ₹18,000 to the usual landlord	known payee, ₹18,000 (20× usual)	None	None	None	Yes	–
L03	First bill to an allow-listed electricity biller	new allow-listed biller, ₹2,450 (3× usual)	None	None	None	Yes	–
L04	₹40,000 deposit to a new landlord, no call	new payee, ₹40,000 (44× usual), ID pasted	Nudge	Nudge	Nudge	Yes	Warns
L05	Father on a call with his son sends ₹500 to son's friend	on a call, new payee, ₹500 (below usual)	None	None	None	Yes	–
L06	₹25,000 first purchase at a new shop, during a call[c]	on a call, new merchant, ₹25,000 (21× usual), Q1 No, Q2 No	Pause	Pause	Pause	Yes	–
L07	₹30,000 to sister (paid before) while on a call with her	on a call, known payee, ₹30,000 (25× usual)	None	None	None	Yes	–
L08	New electrician guides a ₹6,000 payment on a call[c]	on a call, new payee, ₹6,000 (5× usual), Q1 No, Q2 No	Pause	Pause	Pause	Yes	–
L09	Returns ₹2,000 sent by mistake to the person who sent it	on a call, new payee, ₹2,000 (3× usual), ₹2,000 from an unknown sender, pays that sender back	Nudge	Nudge	Nudge	Yes	–
L10	First-time digital user: ₹1,200 at a new shop	new merchant, ₹1,200 (little history)	None	None	None	Yes	–

Stress test: our assumptions, all written down

Assumptions for genuine payments (generator.py)

Assumption	Value
Payee mix: known people / known merchants / billers	22% / 50% / 8%
Payee mix: new people / new merchants / recent (first paid < 7 days ago)	9% / 9% / 2%
On a call while paying: new person / recent / new merchant / known	20% / 15% / 8% / 5–10%
Was on a call in the app in the last 30 minutes	6%
Screen sharing during a payment	0.2%
Payments by people with little history (< 10 in 90 days)	6%
Amount vs own median, new person (lognormal μ, σ)	0.5, 1.2
Already paid other new payees in the last 24 hours	8%
Received money from an unknown sender in 48 h / paid it back	1.5% / 30%
Payee ID pasted: new person / recent / other	40% / 20% / 5%

Results on 2,00,000 genuine sessions (synthetic stress test, base settings)

Rule	Interrupted	Inline Nudge
ScriptBreak v0.3	**6.55 per 1,000**	**3.3%**
Amount rule (first payment to a person, ₹10,000+)	3.73 per 1,000	–

Ablation: what each signal buys and costs (rule pack v0.2, synthetic)

Signal	Scams flagged	False alarms	Kept?
Call seen in app, last 30 min	+5.7 pts	+1.54 per 1,000	Yes
Merchants count as new payees	+9.4 pts	+1.61 per 1,000	Yes
Repeat payment to the same new payee	+0.9 pts	+1.37 per 1,000	No: replaced by adding up
Adding up new-payee payments (last 24 h)	+1.7 pts	+0.85 per 1,000	Yes
3rd new payee within an hour	+0.2 pts	+0.14 per 1,000	Yes
Fixed floor for little history	+3.5 pts	+0.08 per 1,000	Yes
Screen-share + new payee (R2)	+6.9 pts	+0.32 per 1,000	Yes

Results by workflow: rule pack v0.3, synthetic stress test, base settings

Workflow (3,000 sessions each)	Flagged	Critical	Nudge only	Nothing	Amount rule
Digital arrest	**88%**	75%	11%	1%	43%
Fake customer care / refund	**67%**	47%	22%	11%	11%
Fake KYC / bank official	**62%**	47%	20%	18%	8%
Courier / customs	**80%**	62%	16%	3%	37%
AI voice clone of a relative	**76%**	39%	20%	4%	35%
Hang up, then pay (adaptation)	**34%**	23%	62%	4%	56%
"Sent by mistake" return	**100%**	29%	0%	0%	3%
Task scam over chat (out of scope)	**38%**	14%	20%	42%	4%

Same-budget comparison: rule pack v0.3, synthetic stress test, base settings

Rule (amount rules: first payment)	Genuine interrupted	Scams flagged (six in-scope)
ScriptBreak v0.3	**6.6 per 1,000**	**68%**
first payment to a person ≥ ₹10,000	3.7 per 1,000	32%
first payment to a person ≥ ₹7,500	5.7 per 1,000	39%
first payment to anyone ≥ ₹7,500	8.6 per 1,000	44%
first payment to a person ≥ ₹5,000	9.8 per 1,000	49%
first payment to anyone ≥ ₹3,000	29.0 per 1,000	69%

Sensitivity and pre-planned fallback

Genuine calls to new payees vs assumption	0.5×	1×	1.5×	2×	3×
v0.3 interruptions per 1,000	4.5	6.6	8.6	10.7	14.9
With ₹5,000 floor (fallback)	–	4.1	–	6.5	9.1

Fallback: the ≤8 budget is reached at 1.35×; with the ₹5,000 floor, at 2.58×, losing 3 to 15 points of detection across in-scope workflows. **Blanket notice** on every new payee: 20.8% of payments (our Nudge: 3.3%). **Assumed "keep it secret" answers:** 60% of digital-arrest, 50% of voice-clone and courier, 1% of genuine sessions. These numbers test our assumptions, not real-world accuracy.

Pattern-discovery run: every cluster, rule and decision

1,200
synthetic scam reports (labelled as scams, not by type), one month

751
already paused by rule pack v0.2

449
misses clustered (16 event types)

7 of 19
clusters with 20+ misses, covering 390 (87%)

0.45
cut height (silhouette 0.29)

Clusters, largest first

Misses	Mix of true workflows (unknown to the loop)	Rule (or cheapest rule reaching 70%)	Catches	Added per 1,000	Decision
131	hang up and pay 74, chat task 24, courier 12, digital arrest 7, voice clone 7, customer care 4, KYC official 3	unusual amount + left the app and came back + new payee	76%	2.83	Rejected: too costly
73	sent by mistake 73	same amount as money received from an unknown sender + new payee, not paying back that sender	100%	0.05	**Approved: R5**
55	customer care 21, KYC official 12, voice clone 11, courier 9, digital arrest 2	on a call + new payee + payee ID pasted	98%	6.66	Rejected: too costly
43	KYC official 17, customer care 14, courier 6, voice clone 4, digital arrest 2	on a call + new payee	100%	23.83	Rejected: too costly
41	chat task 25, hang up and pay 11, KYC official 3, courier 1, voice clone 1	left the app and came back + new payee + payee ID pasted	78%	6.49	Rejected: too costly
27	chat task 27	left the app and came back + 3rd new payee within an hour + new payee + payee ID pasted	93%	0.02	**Approved: R6**
20	courier 6, customer care 5, KYC official 5, voice clone 3, digital arrest 1	on a call + little payment history + new payee + payee ID pasted	55%	0.37	Shadow mode only

Synthetic held-out test, shifted settings (a different run from appendix B): 2,000 fresh sessions per workflow

Workflow	Digital arrest	Fake customer care / refund	Fake KYC / bank official	Courier / customs	AI voice clone of a relative	Hang up, then pay	"Sent by mistake" return	Task scam over chat
Flagged by v0.2	88%	68%	64%	79%	76%	22%	17%	5%
Flagged by v0.3	**88%**	**68%**	**64%**	**79%**	**76%**	**24%**	**100%**	**27%**

Clustering: average-linkage hierarchical clustering on Jaccard distance between event sets; cut height chosen by best silhouette among cuts that put at least 80% of misses in clusters of 20 or more (a silhouette of 0.29 reflects scams that share tactics; both approved clusters were 100% one workflow). **Rule induction:** conjunctions of the cluster's common events (present in 60%+ of members), plus at most one exclusion, ranked by catches, then false alarms, then simplicity. **Approval policy:** added false alarms at most 0.5 per 1,000 in a 1,00,000-session backtest, at least 70% of the cluster caught, and no genuine behaviour check changed. Approval in this run: the team, applying this written policy. **Genuine interruptions** on 2,00,000 fresh sessions: 6.46 (v0.2) and 6.55 (v0.3) per 1,000. Code: discover.py.

Design review log: changes made after reviewing rule pack v0.1

Two separate AI-assisted adversarial reviews of our first draft (see appendix E) and our own tests led to these changes.

#	Change we made	Why	Where to check
1	Dropped call length; live guidance is a call during the payment, or one seen in the app in the last 30 minutes	Call length can't be read without background monitoring	Slide 6; page 12
2	Added the 30-minute in-app call signal; tested and rejected a stronger rule; we run alongside bank amount rules	"Hang up, then pay" bypassed the call signal	S11, S12; slide 7; page 13
3	Added a fixed ₹5,000 floor for people with little payment history	First-time digital users had no baseline	S13, L10; appendix B
4	Added 24-hour running totals of new-payee payments, and a trigger at the 3rd new payee in an hour	Repeated small payments to one account got only Nudges	S06, S09, S10
5	Counted merchants as new payees; only allow-listed large billers are exempt	A blanket merchant exemption was a bypass	S14, L03, L06
6	Built and ran the discovery loop: 2 rules approved, 1 in shadow mode, 4 rejected	The AI was planned, not shown	Slide 7; appendix C
7	Added held-out tests, written assumptions, a same-budget comparison and a sensitivity analysis with a fallback	Tests were self-graded	Slide 8; appendix B
8	Replaced the "stranger guiding you" question with a secrecy question; added "Ask family" to the Pause screen	Voice-clone victims answered No honestly: Critical rose from 6% to 39% under our assumed answer rates	S07, S16; appendix B
9	Ranked the 48-hour scope Must / Should / Stretch; "Ask family" is Must	The demo relied on a Should item	Slide 10
10	Reworded claims: "flagged", "makes coaching harder", "AI proposes candidate rules", "recovered"	Some claims overreached	Slides 1, 6, 7, 8
11	Matched every caption to the code (known payees, the bank-flag exception, Pause triggers)	Wording contradicted the rules	Slides 5, 6
12	Listed Android, iOS and Google Play limits per input; labelled 30-day and 90-day retention	Platform limits were underplayed	Pages 12, 13
13	No Nudges for known payees or payments under ₹1,000: from 6.5% to 3.3% of payments	Our synthetic stress test showed alert fatigue	Appendix B
14	Removed the "repeat to the same payee" trigger; running totals cover it	Our ablation: it cost more than it caught	Appendix B
15	Cut the main slides to one message each; moved detail to the appendix	Slides were too dense to skim	Slides 6–10, pages 12–13

Disclosures and sources

AI-generated content

Text, page design and code (engine, generator, discovery loop, tests) were drafted with an AI assistant (Claude, by Anthropic). Ideation and the adversarial reviews in appendix D used AI assistants (ChatGPT by OpenAI, and Claude). The team reviewed, edited and can explain every part.

Datasets

No external datasets. All sessions, reports and test cases are synthetic, generated from written assumptions (appendix B) based on public advisories. No bank, customer or platform data. All people, screens, names and UPI IDs shown are fictional.

Models

No model was trained on real data. Decisions use readable rules. Pattern discovery uses hierarchical clustering (SciPy) and a silhouette score (scikit-learn) on synthetic reports.

Third-party assets

Fonts: Inter, Inter Display and Poppins (SIL Open Font License); DejaVu Sans Mono (Bitstream Vera licence). Libraries: NumPy, SciPy, scikit-learn (BSD). The PDF is built with Jinja2 and Playwright. Diagrams and screens were made by the team. Product and organisation names belong to their owners and are used only to describe existing protections.

Derived figure

"4×" on slide 2: ₹4,005.12 crore over 1,03,488 senior-citizen complaints (about ₹3.87 lakh each) against ₹22,495 crore over 24,02,579 complaints overall (about ₹0.94 lakh each), both for 2025 (sources 1 and 2).

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